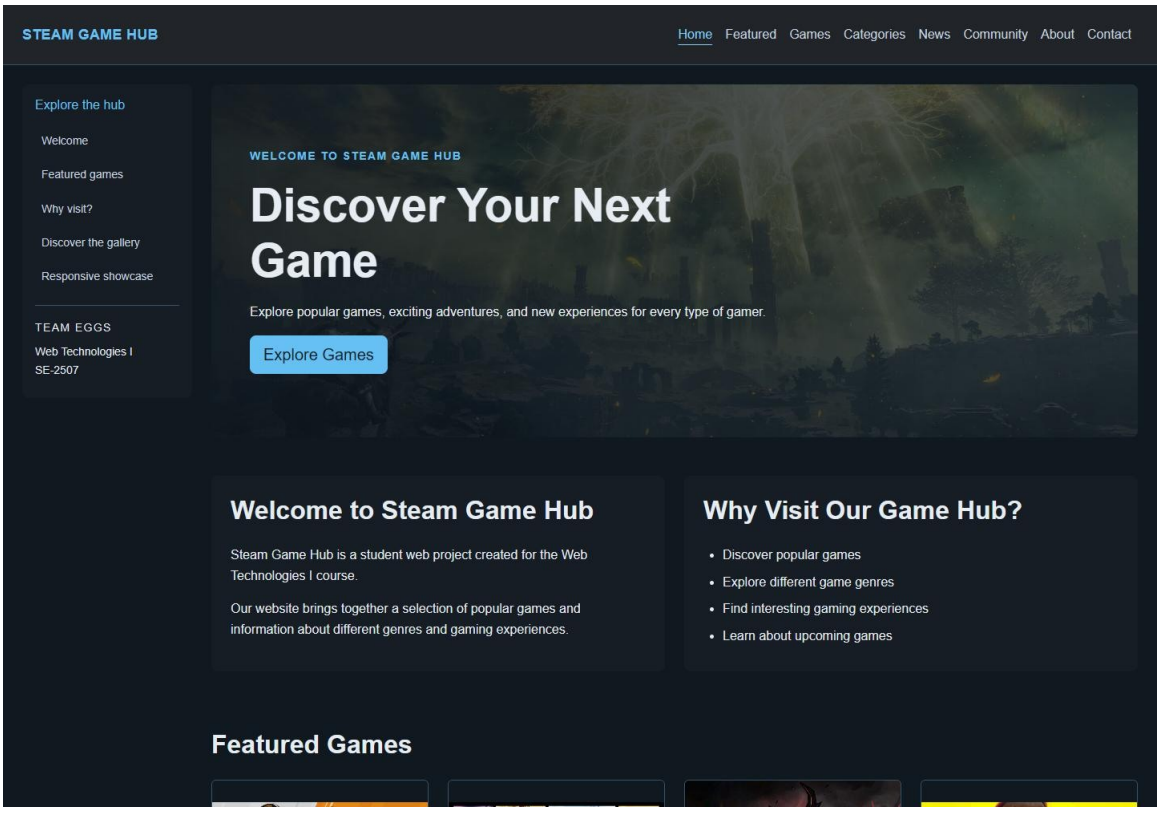
Media Queries + Bootstrap Grid

Assignment 3 · Updated website and implementation report

Team Eggs · Group SE-2507 · Web Technologies I
29 September 2026

Team members
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Objective Extend the existing game hub with responsive media queries, Bootstrap layouts and interactive components while retaining its theme, content and original images.

Deployment status: prepared locally. This updated version has not been deployed; no verified public URL is available. Publishing and LMS submission remain pending.

What is included

Eight Bootstrap pages plus one standalone media-query page. Open index.html directly after extracting the project. All dependencies and images are local.

Task map

Tasks	Location	Main evidence
1-2	responsive.html	Typography and pure-CSS 1 / 2 / 3 cards
3-4	All eight original pages	Bootstrap grid and spacing utilities
5-6	Shared navbar; Games / Featured	Collapse menu, buttons and button group
7-8	About / Community; Home	Nine-image carousel and three-card group
9-10	Contact / News; all pages	Forms, semantics, contrast and keyboard access

Suggested page pairs for the team to review

These pairs organize preparation for the defense. They are not a record of individual authorship; each member should inspect, understand and verify their own work.

Member	Page pair	Components to demonstrate
Нурсултан	Home / Featured	Grid, cards and navigation
Абулхаир	Games / Categories	Grid, buttons and navigation
Шынгыс	News / Community	Forms, carousel and navigation
Артем	About / Contact	Carousel, forms and navigation

All four full names remain in the footer of every HTML page, including the standalone CSS exercise. All 24 original image files are preserved byte for byte.

01 / Responsive typography

responsive.html / css/responsive.css

1. Start with mobile styles: the main heading is 36 px and the introductory paragraph is 16 px.

2. At 768 px, use 52 px and 18 px. At 1200 px, use 68 px and 20 px. The second-level heading changes too.

3. Compare the actual rendered page at 375, 768 and 1440 px; each breakpoint produced the expected font sizes.

Source code

css/responsive.css : lines 15-18

```
15  h1 { font-size: 36px; line-height: 1.08; margin: 16px 0 24px; }
16  h2 { font-size: 24px; line-height: 1.2; margin: 0; }
17  h3 { font-size: 21px; line-height: 1.25; margin: 0 0 12px; }
18  .lead { font-size: 16px; color: #c7d5e0; margin: 0; }
```

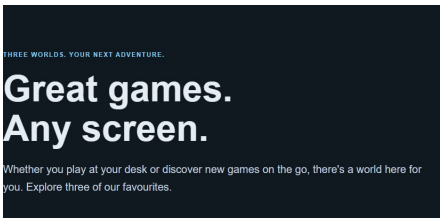
css/responsive.css : lines 36-39

```
36  @media (min-width: 768px) {
37    h1 { font-size: 52px; }
38    h2 { font-size: 28px; }
39    .lead { font-size: 18px; }
```

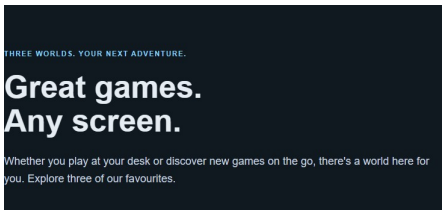
css/responsive.css : lines 43-46

```
43  @media (min-width: 1200px) {
44    h1 { font-size: 68px; }
45    h2 { font-size: 32px; }
46    .lead { font-size: 20px; }
```

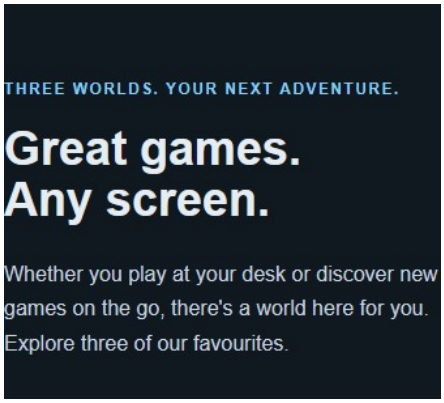
Rendered result



Desktop: 68 px



Tablet: 52 px



Mobile: 36 px

02 / Three cards with media queries

responsive.html / css/responsive.css

1. Create exactly three articles with an image, heading, description and a working game link.

2. Use a one-column CSS Grid by default, two columns at 768 px and three at 1200 px.

3. Load only responsive.css on this page. No Bootstrap CSS or JavaScript can affect this exercise.

Source code

css/responsive.css : lines 23-23

```
23  .mq-cards { display: grid; grid-template-columns: minmax(0, 1fr); gap: 24px; }
```

css/responsive.css : lines 41-41

```
41  .mq-cards { grid-template-columns: repeat(2, minmax(0, 1fr)); }
```

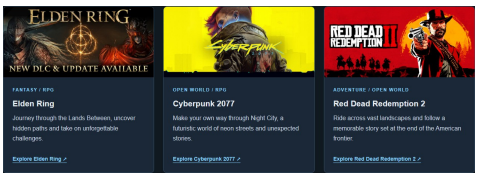
css/responsive.css : lines 48-48

```
48  .mq-cards { grid-template-columns: repeat(3, minmax(0, 1fr)); }
```

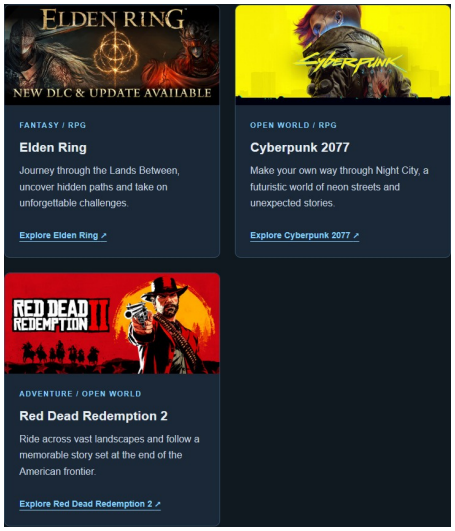
responsive.html : lines 8-9

```
8  <!-- Tasks 1-2 intentionally use no Bootstrap stylesheet or script. -->
9  <link rel="stylesheet" href="css/responsive.css">
```

Rendered result



3 columns / 1440 px



2 columns / 768 px



1 column / 375 px

03 / Bootstrap grid layout

index.html / games.html

1. Wrap the site in container-fluid, then use row and gutter classes for layout.

2. Place the Home introduction and benefits in two col-lg-6 columns. The catalogue uses col-12 col-md-6 col-lg-4.

3. The same Bootstrap grid also lays out news, genres, team members, the contact form and carousel containers.

Source code

index.html : lines 146-149

```
146  <div class="row g-4 my-4" id="hub-overview">
147   <div class="col-12 col-lg-6">
148    <section class="h-100 rounded-3 p-3 p-md-4 surface-panel" id="welcome">
149     <h2 class="mb-4 fw-bold">
```

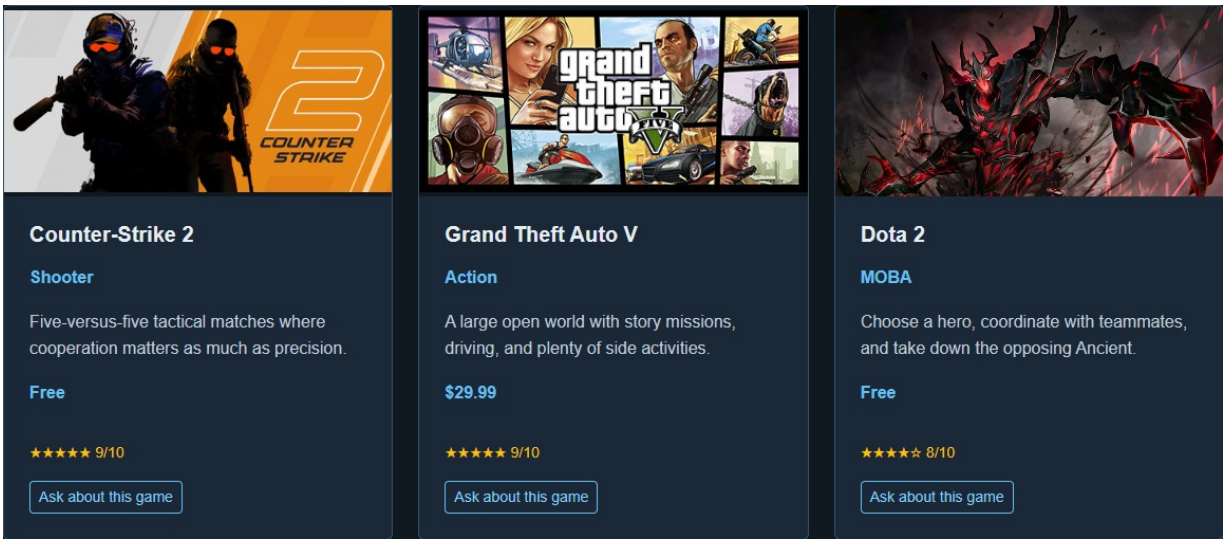
games.html : lines 156-160

```
156  <div class="col-12 col-md-6 col-lg-4">
157   <article class="game-card card h-100" id="counter-strike-2">
158    <img alt="Counter-Strike 2 - official Steam cover artwork" class="card-img-top" height="215"
     src="images/games/counter-strike-2.jpg" width="460"/>
159    <div class="card-content card-body d-flex flex-column p-3 p-lg-4">
160     <h3 class="h5 mb-3 fw-bold card-title">
```

Rendered result



Home: two columns



Games: three columns

04 / Bootstrap spacing utilities

All eight Bootstrap pages

1. Remove the previous layout stylesheets and apply Bootstrap margin, padding and gutter classes in HTML.

2. Use p-3, p-md-4, py-lg-5, g-4 and responsive margins such as mt-lg-5 to create consistent spacing.

3. Keep theme.css focused on appearance. It contains no custom margin or padding declarations. The pure-CSS exercise is separate.

Source code

index.html : lines 81-85

```
 81  <div class="container-fluid px-3 px-sm-4 py-4 flex-grow-1">
 82   <div class="row g-4">
 83    <aside aria-labelledby="sidebar-title" class="page-sidebar col-12 col-xl-2">
 84     <div class="sidebar-panel rounded-3 p-3">
 85      <h2 class="h6 text-info mb-3" id="sidebar-title">
```

games.html : lines 317-321

```
317  <nav aria-label="Browse games" class="mt-4 mt-lg-5">
318   <div aria-label="Game collections" class="btn-group" role="group">
319    <a class="btn btn-primary btn-sm" href="featured.html">
320     Featured
321    </a>
```

Rendered result



Responsive spacing in the Home overview

05 / Collapsible Bootstrap navigation

Navbar shared by eight pages

1. Use a semantic nav with eight links and the navbar, navbar-expand-xl and bg-dark classes.

2. Connect navbar-toggler to main-nav using data-bs-target and aria-controls. Load the local Bootstrap bundle.

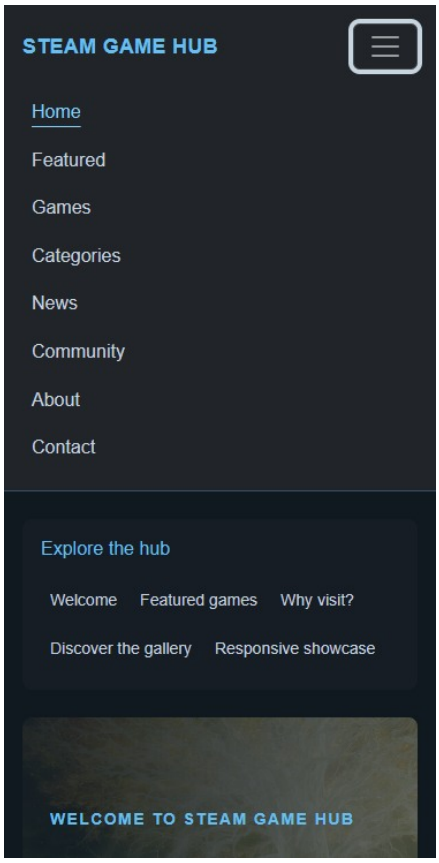
3. Below 1200 px, open and close the hamburger menu. Confirm aria-expanded changes and each page can navigate to Contact.

Source code

index.html : lines 24-33

```
24  <nav aria-label="Main navigation" class="navbar navbar-expand-xl bg-dark border-bottom py-3">
25   <!-- Tasks 4-5: Bootstrap spacing and Collapse navbar. -->
26   <div class="container-fluid px-3 px-sm-4">
27    <a class="navbar-brand fw-bold text-info me-2" href="index.html">
28     STEAM GAME HUB
29    </a>
30    <button aria-controls="main-nav" aria-expanded="false" aria-label="Toggle navigation" class="navbar-
    toggler" data-bs-target="#main-nav" data-bs-toggle="collapse" type="button">
31     <span class="navbar-toggler-icon">
32     </span>
33    </button>
```

Rendered result



375 px: menu expanded

06 / Bootstrap buttons and groups

Games / Featured / shared components

1. Replace the old main-button and secondary-button styles with btn-primary and btn-outline-info.

2. Use btn-lg for primary calls to action and btn-sm for card links and collection controls.

3. Use btn-group with an accessible group label. Featured, All games and Genres link to real pages.

Source code

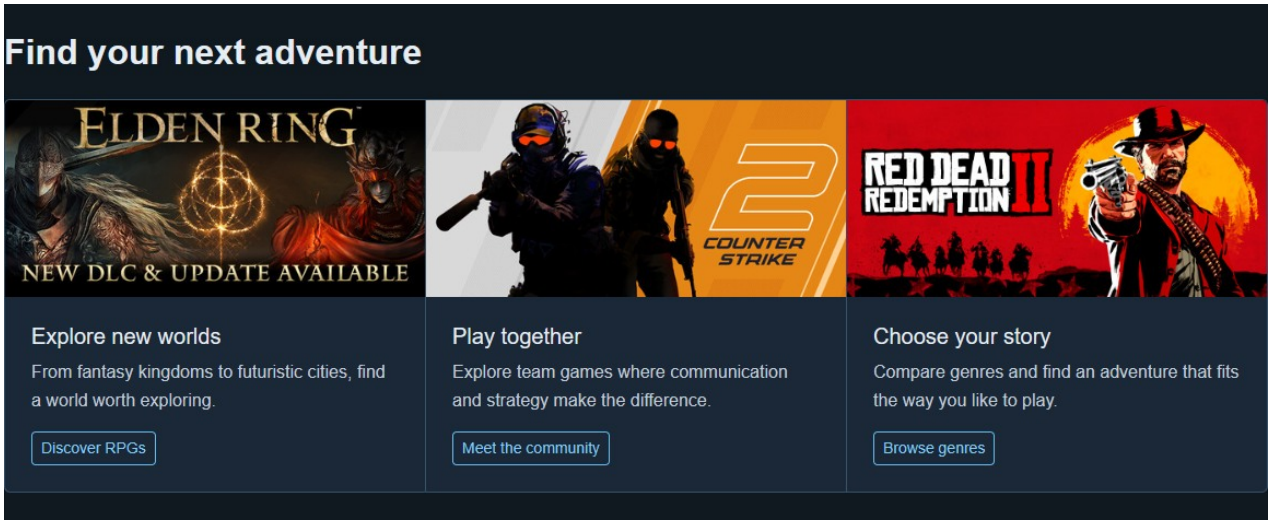
games.html : lines 318-330

```
318    <div aria-label="Game collections" class="btn-group" role="group">
319     <a class="btn btn-primary btn-sm" href="featured.html">
320      Featured
321     </a>
322     <a class="btn btn-outline-info btn-sm" href="games.html">
323      All games
324     </a>
325     <a class="btn btn-outline-info btn-sm" href="categories.html">
326      Genres
327     </a>
328    </div>
329   </nav>
330  </section>
```

Rendered result



Related collection controls



Small outline buttons on cards

07 / Nine-image Bootstrap carousel

About / Community

1. Convert each nine-image gallery into carousel-inner and nine carousel-item elements, preserving every source image.

2. Add nine indicators plus previous and next controls. Keep the carousel in a Bootstrap column.

3. Disable automatic cycling. Verify all nine images, wrapping, indicators, arrow keys and the live slide announcement.

Source code

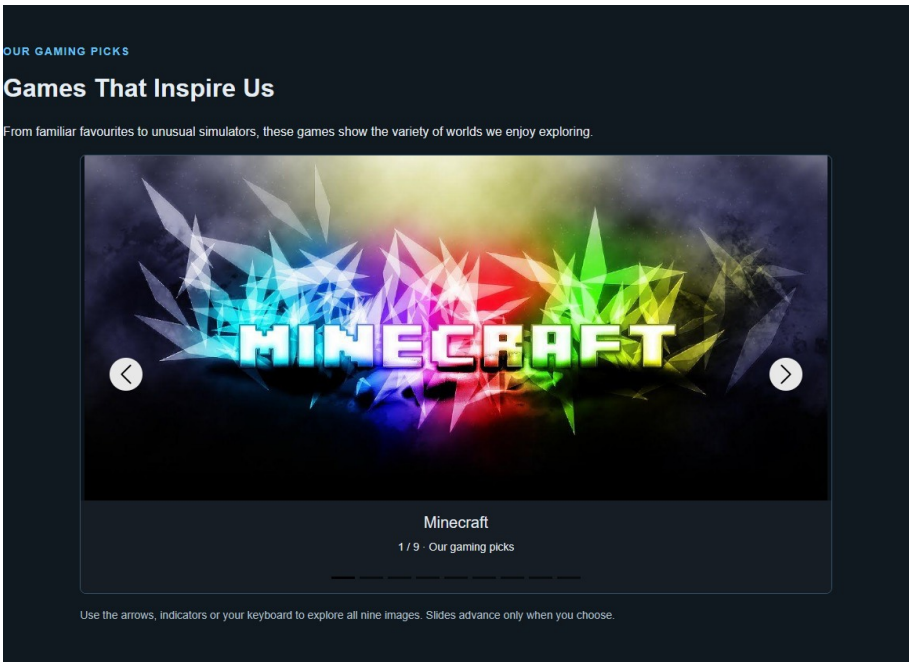
about.html : lines 215-219

```
 215  <div aria-label="Game gallery" aria-roledescription="carousel" class="carousel slide rounded-3
      overflow-hidden border" data-bs-interval="false" id="games-carousel" role="region" tabindex="0">
 216   <div class="carousel-indicators mb-2">
 217    <button aria-current="true" aria-label="Slide 1: Minecraft" class="active" data-bs-slide-to="0" data-
      bs-target="#games-carousel" type="button">
 218    </button>
 219    <button aria-label="Slide 2: Dota 2" class="" data-bs-slide-to="1" data-bs-target="#games-carousel"
      type="button">
```

about.html : lines 337-343

```
 337  <button class="carousel-control-prev" data-bs-slide="prev" data-bs-target="#games-carousel"
      type="button">
 338   <span aria-hidden="true" class="carousel-control-prev-icon">
 339   </span>
 340   <span class="visually-hidden">
 341    Previous image
 342   </span>
 343  </button>
```

Rendered result



About: original images in a manually controlled carousel

08 / Bootstrap cards

Home / discover-more

1. Create a card-group of three Bootstrap cards. Each contains an image, a title and a description.

2. Use card-body, card-title, card-text and Bootstrap buttons. Flex utilities keep actions aligned at the bottom.

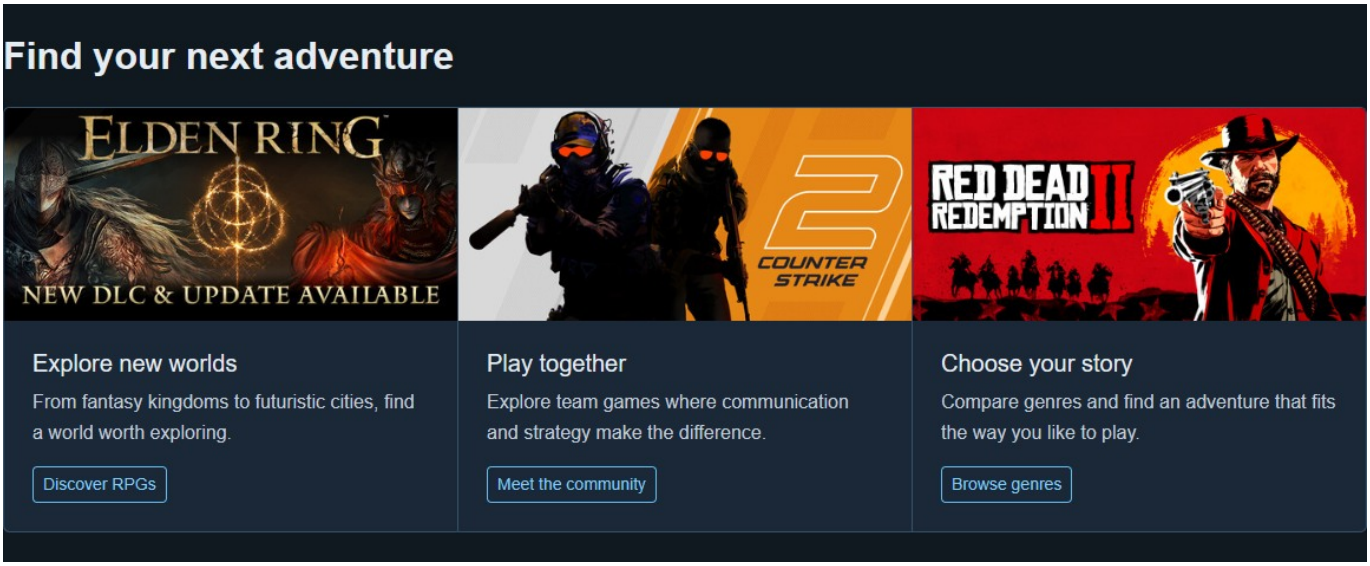
3. Retain additional Bootstrap cards on the catalogue, news, genres and team pages.

Source code

index.html : lines 282-296

```
282  <div class="card-group">
283   <article class="card">
284    <img alt="Elden Ring official cover" class="card-img-top game-cover" src="images/games/elden-
     ring.jpg"/>
285    <div class="card-body p-3 p-lg-4 d-flex flex-column">
286     <h3 class="card-title h5">
287      Explore new worlds
288     </h3>
289     <p class="card-text">
290      From fantasy kingdoms to futuristic cities, find a world worth exploring.
291     </p>
292     <a class="btn btn-outline-info btn-sm mt-auto align-self-start" href="featured.html#elden-ring">
293      Discover RPGs
294     </a>
295    </div>
296   </article>
```

Rendered result



Three complete cards inside card-group

09 / Responsive Bootstrap forms

Contact / News

1. Use form-label, form-control, form-select and input-group. Add correctly labelled radio buttons and a checkbox.

2. Arrange contact information and the form as col-12 col-lg-6 columns. Inputs use native browser validation.

3. Check empty and valid submissions. Show an aria-live result locally; the demonstration sends and stores no information.

Source code

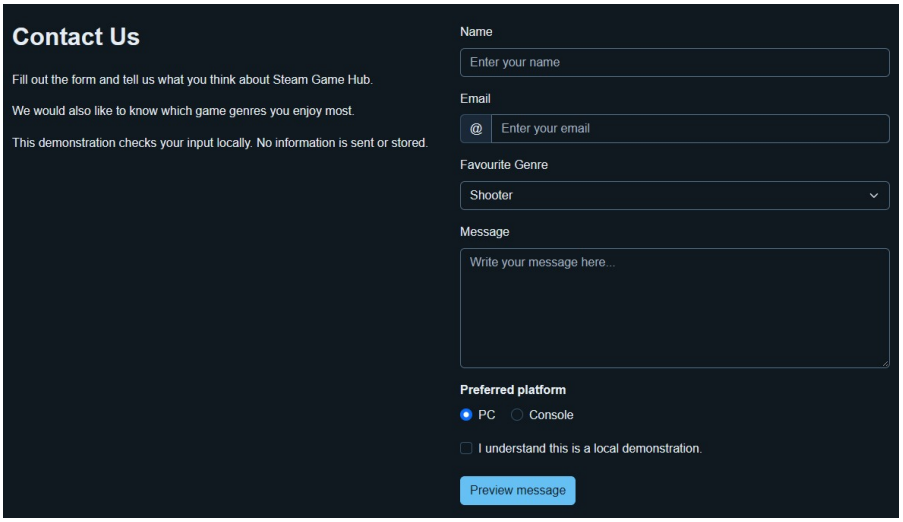
contact.html : lines 163-167

```
163  <div class="input-group">
164   <span aria-hidden="true" class="input-group-text">
165    @
166   </span>
167   <input autocomplete="email" class="form-control" id="email" name="email" placeholder="Enter your
     email" required="" type="email"/>
```

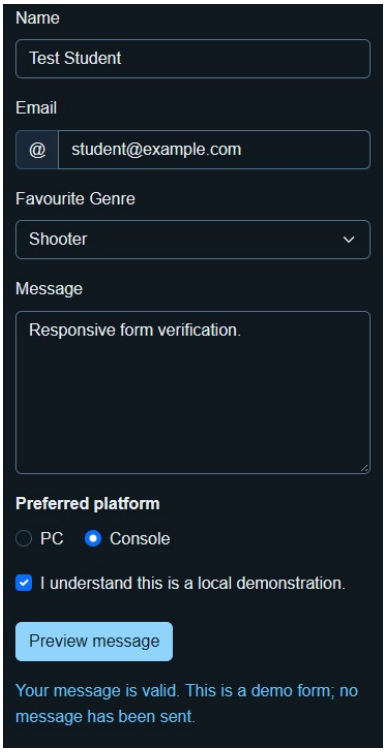
contact.html : lines 202-208

```
202  <div class="form-check form-check-inline">
203   <input checked="" class="form-check-input" id="platform-pc" name="platform" type="radio" value="pc"/>
204   <label class="form-check-label" for="platform-pc">
205    PC
206   </label>
207  </div>
208  <div class="form-check form-check-inline">
```

Rendered result



Desktop: two-column contact layout



Mobile: validated form

10 / Accessibility and semantics

All nine pages

1. Retain header, nav, main and footer landmarks; add useful image alternatives, visible labels and a keyboard skip link.

2. Provide high-contrast focus outlines and reduced-motion styles. Make the horizontally scrollable game table keyboard focusable.

3. Run automated WCAG A/AA checks at mobile and desktop sizes, then verify focus, navigation and carousel keyboard operation.

Source code

index.html : lines 20-22

```
20  <a class="visually-hidden-focusable btn btn-light position-absolute top-0 start-0 m-2 skip-link"
    href="#main-content">
21   Skip to main content
22  </a>
```

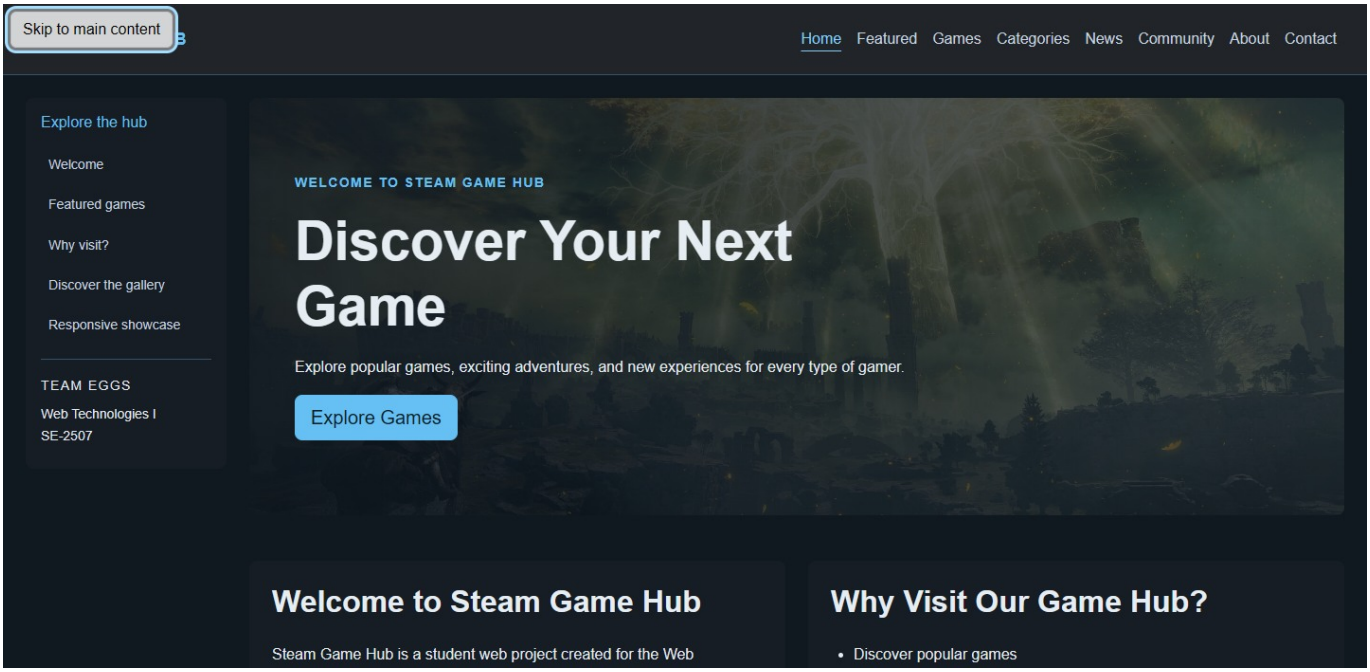
games.html : lines 335-336

```
335  <div aria-label="Game comparison table; scroll horizontally on small screens" class="table-responsive
     rounded-3 border" role="region" tabindex="0">
336   <table class="games-table table table-dark table-hover align-middle mb-0">
```

css/theme.css : lines 68-69

```
68  :focus-visible { outline: 3px solid #a5dfff; outline-offset: 3px; }
69  .btn:focus-visible, .navbar-toggler:focus-visible { outline: 3px solid #a5dfff; outline-offset: 3px; }
```

Rendered result



Visible keyboard focus on the skip link

Verification and next steps

81 / 81 responsive checks passed. Nine pages were opened at 320, 375, 576, 767, 768, 992, 1199, 1200 and 1440 px in headless Microsoft Edge (154.0.4258.37).

• No horizontal page overflow, broken images, duplicate IDs, unlabelled inputs or JavaScript page errors were detected.

• All 271 local references resolve, including page anchors. All 24 source images match the original files. Bootstrap files match the official SHA-384 values.

• Eight navbar tests verified opening, closing, aria-expanded and navigation. Both carousels displayed all nine images, wrapped and responded to indicators and the keyboard.

• Both forms produced local success messages. Empty required contact fields were blocked. The skip link moved keyboard focus to main.

• Automated axe-core WCAG A/AA checks returned zero violations on 18 page/viewport combinations after fixing keyboard access to the mobile comparison table. Automated checks do not replace a complete manual accessibility audit.

Reflection

Bootstrap makes the repeated navigation, grid, cards and forms consistent across pages. Keeping the media-query exercise separate makes its 1 / 2 / 3-column behavior easy to explain. The main integration risks were old layout rules overriding Bootstrap and interactive elements lacking keyboard support; the old stylesheets were replaced and those interactions were checked.

Part 3 / publication remains pending

No commit, push, deployment or LMS submission was performed. Publish this project through the team repository using GitHub Pages or Netlify, verify the live site and add its URL to the report. Each member must submit the required files and prepare for the classroom defense.

Sources and evidence

Assignment brief: Web Assignment 3 Media Queries + Bootstrap Grid.pdf.
Bootstrap 5.3 documentation: Getting started, Cards, Carousel.
Image provenance: IMAGE_SOURCES.md. Full screenshots and source excerpts: report/screenshots/. Machine-readable results: report/verification.json.